

Living Off the Land

Scheduled Tasks

CIS Critical Security Controls v8

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Acknowledgements

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Introduction

62% of attackers use LotL tools or techniques in their attacks

This guide provides an overview of a well-known Living off the Land (LotL) technique called Scheduled Tasks—what it is, how it is abused, and how you can shore up defenses for this type of attack.

IT administrators and security professionals will find this content helpful within their own cybersecurity programs in taking a defense-in-depth approach to defending against LotL attacks involving scheduled tasks.

For additional information on the CIS Controls, visit <https://www.cisecurity.org/controls>

Living off the Land (LotL) attacks involve the use of existing tools and tactics on targeted systems or networks to carry out a cyber attack, rendering the attack difficult to detect and defend. According to CrowdStrike's 2022 Global Threat Report, 62% of attackers are using LotL tools or techniques in their attacks.¹ There are several techniques that an attacker can use, but some of the top techniques used include PowerShell, Windows Command Shell, Windows Management Instrumentation (WMI), and Scheduled Tasks/Jobs.² Why do attackers use LotL techniques? It's simple. These techniques are easily accessible, can evade detection, are highly adaptive, and can often be automated.

CIS has published several guides to provide enterprises with specific guidance defending against many of these LotL techniques,³ including this most recent guide: *Living off the Land: Scheduled Tasks*. Scheduled tasks are a common technique used by attackers to automate and perform malicious activities on target systems. In fact, Scheduled Task (T1053.005) was the seventh most prevalent LotL ATT&CK® sub-technique used in 2022.⁴ Attackers use scheduled tasks for a variety of reasons including at startup to initiate an infection, on a recurring basis to establish persistence on a system, or for lateral movement to spread from system to system. Unfortunately, a “block all scheduled tasks” approach is not realistic. System Administrators often use scheduled tasks to automate activities across the network so that tedious and time-consuming tasks are more manageable.

The Center for Internet Security® (CIS) is the home of the CIS Critical Security Controls® (CIS Controls®), well-regarded and widely used best practice recommendations that help enterprises focus their resources on the most critical defensive actions. In keeping with the mission of the CIS Controls, we continue to provide prioritized, simplified, and relevant guidance to defend against the most common threats that plague enterprises. This guide, *Living off the Land: Scheduled Tasks*, aims to do just that by providing readers with an overview of how this tool is legitimately used, how it is abused, and what enterprises can do to protect against an attack that may use scheduled tasks.

¹ <https://www.crowdstrike.com/blog/why-you-need-ai-and-machine-learning-to-combat-hands-on-keyboard-attacks/>

² As reported in Red Canary's 2022 Threat Detection Report of the top MITRE ATT&CK Techniques (https://resource.redcanary.com/rs/003-YRU-314/images/2022_ThreatDetectionReport_RedCanary.pdf)

³ *Living off the Land Attacks: PowerShell, Commonly Exploited Protocols: Windows Management Instrumentation, Exploited Protocols: Server Message Block (SMB) for CIS Controls v8, and Exploited Protocols: Remote Desktop Protocol (RDP) for CIS Controls v7.1.*

⁴ Based on Red Canary's 2022 Threat Detection Report of top ATT&CK (Adversarial Tactics, Techniques, and Common Knowledge) (Sub-) Techniques <https://redcanary.com/threat-detection-report/techniques/scheduled-task/>

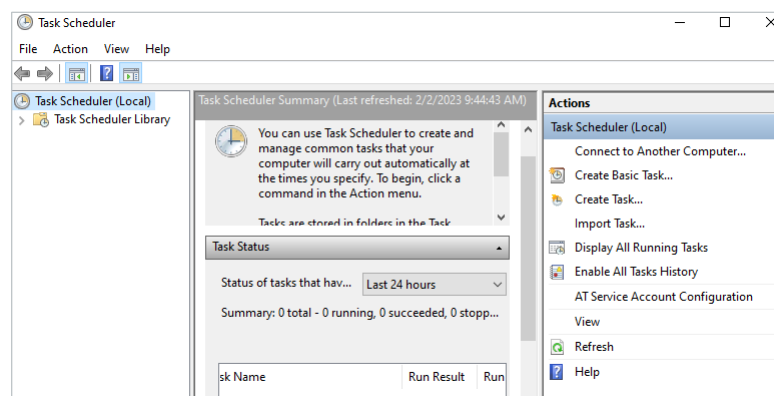
Scheduled Tasks

What Are Scheduled Tasks?

Scheduled tasks are a legitimate tool used by administrators for the automation of processes. They can be set to run at specific times, can run based on a triggering event, or can be run repeatedly or one-time. The Windows® Task Scheduler, a graphical user interface (GUI) of the Windows® operating system (OS), is the application that allows scheduled tasks to run. It can be used to run scripts, launch applications, or perform other tasks—both locally and remotely from another system (Figure 1).

While scheduled tasks can be used on multiple operating systems (e.g., cron jobs on Linux® and macOS®⁵), this guide will focus on scheduled tasks for the Windows OS.

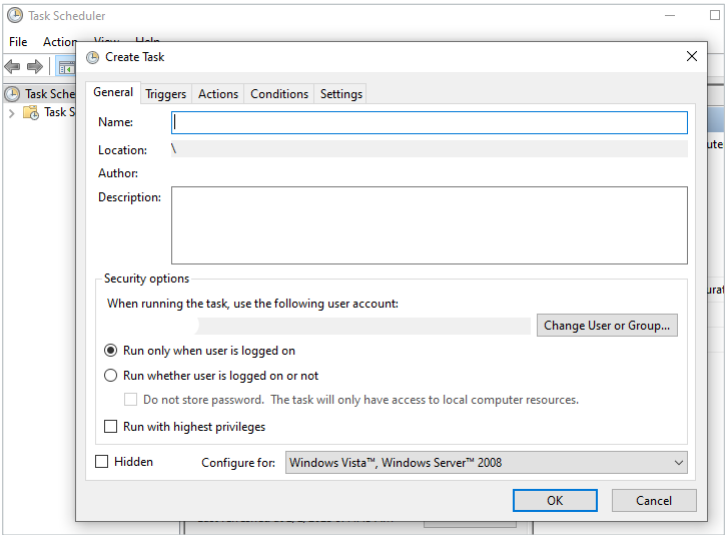
Figure 1. Windows Task Scheduler



Within the GUI, there are a variety of options when creating tasks, including running with the highest privileges, hiding a scheduled task, and even changing the user or group that it runs under. While these sound like particularly helpful features, keep in mind that attackers may also abuse these features if they gain access to the system or network (Figure 2).

⁵ Linux® is the registered trademark of Linus Torvalds in the U.S. and other countries. macOS® is a trademark of Apple Inc., registered in the U.S. and other countries.

Figure 2. Windows Task Scheduler: Creating a Task



There are also several other options for scheduling tasks, including the ability to add a date or time trigger on a recurring basis (Figure 3), the action under which the task will perform (e.g., starting a program) (Figure 4), and the conditions in which the task will run (e.g., based on the state of the computer: idle, connected to power and/or network) (Figure 5), as shown below.

Figure 3. Windows Task Scheduler: Triggers (Left)

Figure 4. Windows Task Scheduler: Actions (Right)

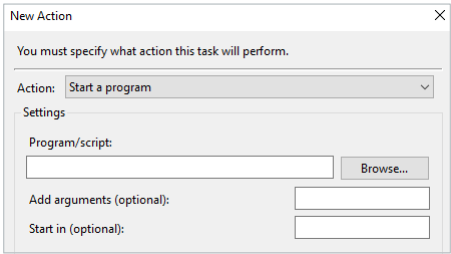
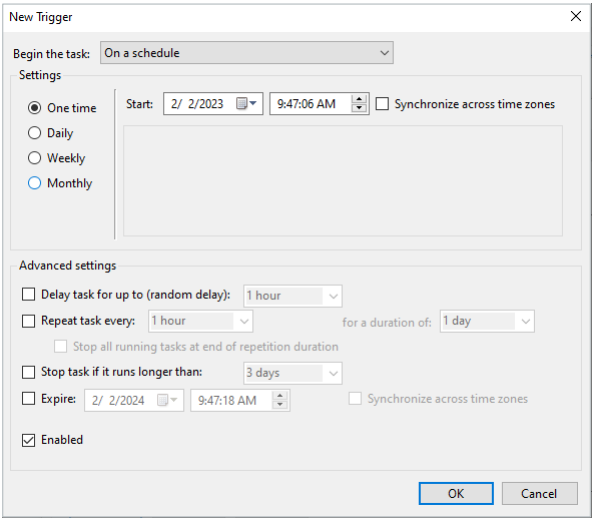
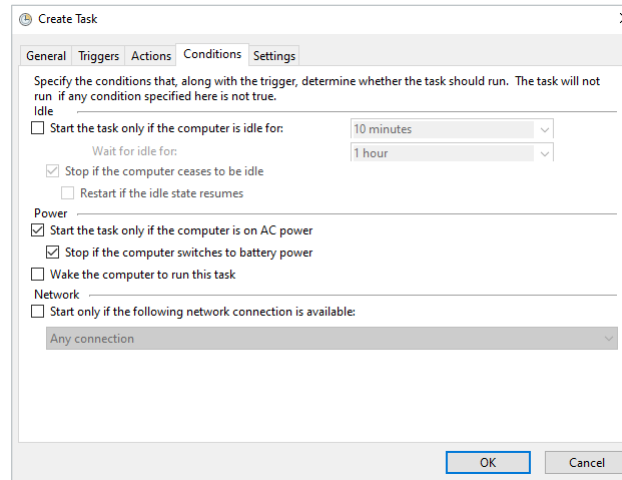
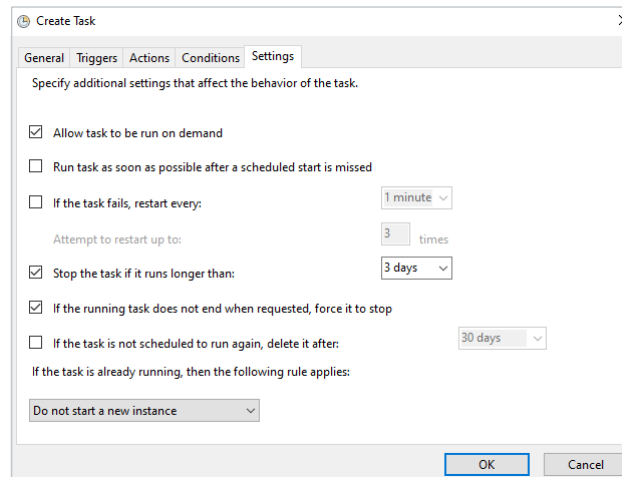


Figure 5. Windows Task Scheduler: Conditions



Additionally, Task Scheduler gives the option to perform actions such as running the task on demand, restarting the task if it fails, stopping the task after a set period of time, forcing the task to stop, and even deleting the task if it is no longer scheduled to run (Figure 6). Again, these are all great features, but are also ones that can be abused if given the opportunity.

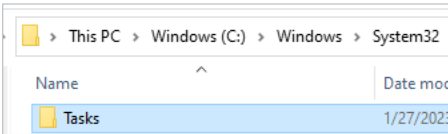
Figure 6. Windows Task Scheduler: Settings



Administrators may also use Task Scheduler via the command line interface (CLI) with the "schtasks" command to run programs or scripts at a scheduled date and time. Another method, which has since been deprecated as of Windows 8, is using the "at.exe" command. This command-line utility allows a user to schedule a one-time task to run at a specific time or date and can be run with elevated privileges.

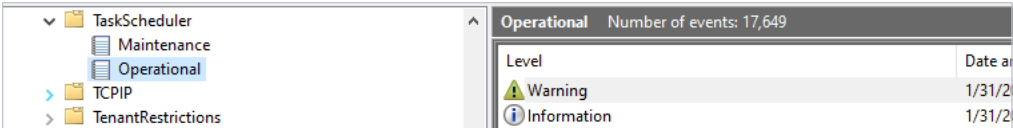
Once a scheduled task is created, it can be found in the “%SystemRoot%\Windows\System32\Tasks” directory of the Windows OS, as shown below in Figure 7.

Figure 7. File Directory for Scheduled Tasks



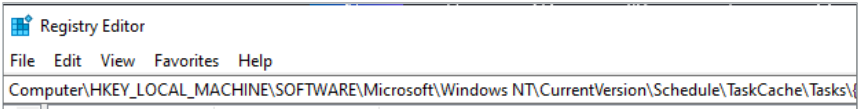
As with many other actions performed on the Windows OS, events are generated and stored in event logs for storage and review. Events associated with scheduled tasks are logged within the Windows Task Scheduler Operational Event Log (“%SystemRoot%\Windows\System32\winevt\Logs\Microsoft-Windows-TaskScheduler%4Operational.evtx”), as shown below in Figure 8.

Figure 8. Windows Task Scheduler Operational Event Log



Another important item to note is the location of the scheduled tasks in the Windows Registry. Shown below in Figure 9 is the location of the registry key for scheduled tasks, located in the SOFTWARE registry hive. This information will become important later as we talk about how attackers can abuse this and use it to evade detection.

Figure 9. Windows Registry Key for Scheduled Tasks



Benefits of Using Scheduled Tasks

There are several benefits of using scheduled tasks, including:

- **Automation:** Scheduled tasks have the ability to run on a pre-defined schedule, freeing up time that would otherwise be used to manually perform these actions.
- **Efficiency:** Automating tasks increases not only the efficiency of the system, but also of the administrators who are responsible for performing the tasks.
- **Consistency:** Scheduling tasks on a recurring basis reduces the possibility for error as well as places responsibility on the system for ensuring that the task is run regularly – rather than relying on a human to remember to perform the task.
- **Cost Savings:** By automating scheduled tasks, the cost of labor is reduced or repurposed to perform other activities that may not have the ability to be automated.

Attacks Abusing Scheduled Tasks

Part of defending against a LotL attack is knowing how the tool operates normally within an environment, as we discussed above. This helps enterprises understand what is normal, and begin to establish a baseline in systems to identify when activity is abnormal. Scheduled task techniques vary across several tactics of the MITRE ATT&CK® framework including execution, persistence, and privilege escalation. Because of this, a defender's approach to defending against attacks involving LotL tools must be multi-faceted.

There are several ways an attacker can abuse scheduled tasks. Many of them are legitimate use cases listed above, while others are more obvious in their malicious intent, including:

- Using scheduled tasks to initiate an infection
- Setting up a scheduled task under the SYSTEM account to run at startup as a method of persistence as well as for lateral movement
- Bypassing User Access Control (UAC) through Task Scheduler by selecting "Run with highest privileges" (Figure 2)
- Evading detection by disguising the process as trusted or signed
- Evading detection by deleting the registry value of the Security Descriptor (SD) using SYSTEM privileges

Additionally, attackers can use a variety of parameters with the "schtasks" command including: change, delete, end, query, run, or create. These parameters, while appearing fairly benign, can give attackers a lot of power to cause damage to a system or network. A description of these parameters is shown below in Figure 10.

Figure 10. "Schtasks" Parameters

Image Source: <https://learn.microsoft.com/en-us/windows-server/administration/windows-commands/schtasks>

Parameter	Description
<code>schtasks change</code>	Changes one or more of the following properties of a task: • The program that the task runs (/tr) • The user account under which the task runs (/ru) • The password for the user account (/rp) • Adds the interactive-only property to the task (/it)
<code>schtasks create</code>	Schedules a new task.
<code>schtasks delete</code>	Deletes a scheduled task.
<code>schtasks end</code>	Stops a program started by a task.
<code>schtasks query</code>	Displays tasks scheduled to run on the computer.
<code>schtasks run</code>	Starts a scheduled task immediately. The <code>run</code> operation ignores the schedule, but uses the program file location, user account, and password saved in the task to run the task immediately.

The use of scheduled tasks is widely abused by many threat actor groups and is incorporated into several malware variants. Some more well-known malware variants, past and current, that use scheduled tasks in their attacks include: Tarrask (used by HAFNIUM), RedLine, Agent Tesla, Emotet, IcedID, Ryuk, TrickBot, and others.

While it might seem like there is no hope in defending against a LotL attack, there are defensive steps that an enterprise can take to secure their environment and help to mitigate the success of an attack. The CIS Controls offer a set of best practice recommendations that work together to develop a defense-in-depth approach to cybersecurity. Additionally, our CIS Benchmarks™ offer secure configuration guidelines for 100+ technologies, including operating systems, applications, and network devices. Combined, they form a formidable defense against a scheduled task attack.

Scheduled Task Defenses

Defending against a LotL attack does not need to be a reactive process. With any good cybersecurity program, the first step in implementing defenses against any attack is to know your environment. In order to defend a network, you must first know what is on the network. This includes enterprise assets (e.g., workstations, servers), software, and data. Without taking this first step in your cybersecurity program, it is difficult to implement additional Safeguards. This includes Implementation Group 1 (IG1) Safeguards that are a part of CIS Controls 1, 2, and 3.⁶

Once these foundational controls are implemented, there are several ways to proactively defend against an attack that abuses scheduled tasks. Below are six simple actions that enterprises can take to strengthen their defenses against a scheduled task attack. As with any best practice, test your changes before pushing them out to production, as configurations are not a "one size fits all" activity. Enterprises may require slightly different configuration settings that are most appropriate for their environment.

Remove/Limit Access to "schtasks.exe"

Privileged account management is one way to reduce the chance of attackers gaining access to higher privileged accounts that can be used to abuse scheduled tasks. Generally speaking, information technology (IT) administrators, or similar positions, will require access to perform various actions (e.g., create, modify, delete) on scheduled tasks. However, regular users will likely not need the ability to perform these actions. To reduce this risk, enterprises can safely remove or limit the ability to perform these scheduled task functions to only those who have a business need for it (e.g., administrators).

This is particularly important when it comes to accessing a system remotely, as attackers can and often will schedule tasks remotely from another system on or outside of the network, depending on account permissions and system configurations. Additionally, reducing the ability for regular user accounts to escalate privileges is also important. This will help mitigate the potential for attackers to abuse administrator accounts by way of compromising a regular user account.

A Note About Protecting Privileged Accounts

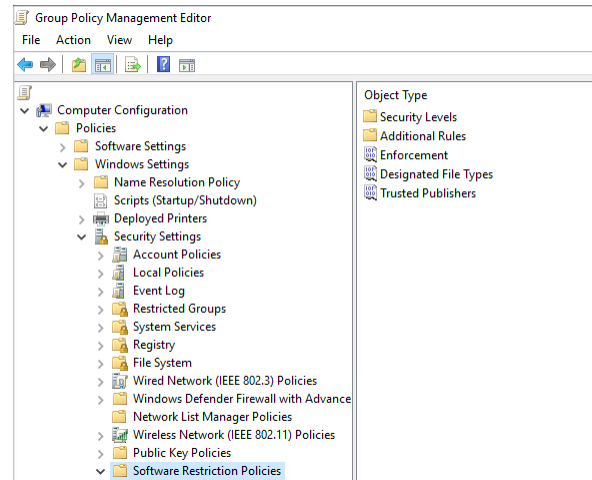
Attackers do not discriminate. Wherever there is an account to be compromised, created, or escalated to higher privileges, they will take advantage. Administrator accounts, or those with higher privileges, are a particular target. These types of accounts allow attackers to perform malicious activities such as creating other accounts, changing configuration settings, or exploiting vulnerabilities.

It is worth noting that simply limiting the functionality of scheduling tasks to administrator accounts only is not enough. Administrator accounts require additional protections, such as restricting administrator privileges to dedicated administrator accounts. This means that conducting general computing activities, such as internet browsing, email, and productivity suite use, from the user's primary, non-privileged account, and any administrative activities from the user's administrator account. Additionally, if the account allows for it, using multi-factor authentication (MFA) can help to reduce the chance of an account compromise. At a minimum, using unique passwords will help to reduce the risk. However, none of these actions are effective individually. Taking a defense-in-depth approach to defending against any LotL attack is necessary to reduce the chance of becoming a victim.

⁶ CIS Control 1: Inventory and Control of Enterprise Assets, CIS Control 2: Inventory and Control of Software Assets, and CIS Control 3: Data Protection

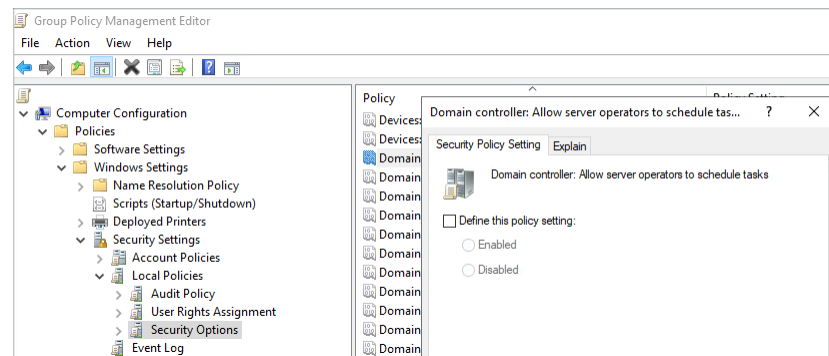
A common method to restrict access is through Software Restriction Policies in Windows. This can be accomplished in Group Policy by blocking the “schtasks.exe” executable from running. This Group Policy setting can be found under “Computer Configuration\Policies\Windows Settings\Security Settings\Software Restriction Policies” (Figure 11).

Figure 11. Configuration Setting: “Computer Configuration\Policies\Windows Settings\Security Settings\Software Restriction Policies”



Another way that attackers can abuse “schtasks.exe” is by having the scheduled task run under the authenticated user account instead of running under an account with higher privileges (e.g., the SYSTEM account). This setting can be found within the Registry key: “HKLM\SYSTEM\CurrentControlSet\Control\Lsa\SubmitControl.” This can be accomplished by configuring the policy setting ‘Domain Controller: Allow server operators to schedule tasks’ to ‘Disabled,’ located in “Computer Configuration\Policies\Windows Settings\Security Settings\Local Policies\Security Options” (Figure 12).

Figure 12. Configuration Setting: “Computer Configuration\Policies\Windows Settings\Security Settings\Local Policies\Security Options”

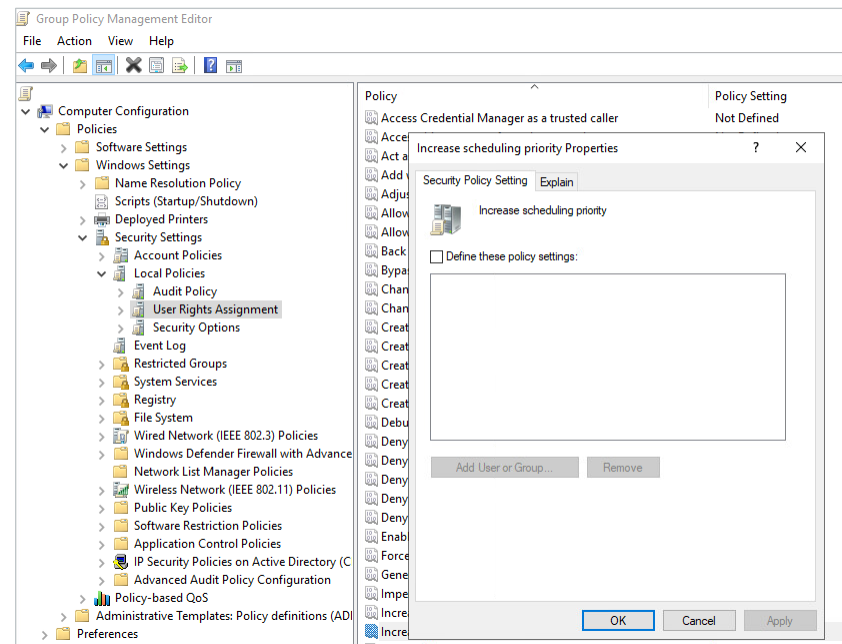


Reduce the Ability to Raise the Priority of a Scheduled Task

When a scheduled task is created, it is assigned a priority. This priority can be modified, authorized or unauthorized, to change it to a higher or lower priority. If a user, who is given this permission, increases the priority level of a process (e.g., to Real-Time), this would leave little processing time for all other processes and could lead to a Denial of Service (DoS) attack.

To reduce the chance of an attacker modifying the priority of a scheduled task to a higher level, a Group Policy setting can be configured, under “Computer Configuration\Policies\Windows Settings\Security Settings\Local Policies\User Rights Assignment\Increase scheduling priority,” to allow only administrators to change this priority level (CIS Windows 10 Enterprise Benchmark 2.2.25) (Figure 13).

Figure 13. Configuration Setting: “Computer Configuration\Policies\Windows Settings\Security Settings\Local Policies\User Rights Assignment\Increase scheduling priority”



Restrict Access to “at.exe”

“At.exe” is a legitimate executable that was developed by Microsoft® to run scheduled tasks. However, it has since been deprecated as of Windows 8. While “at.exe” cannot access any tasks that are created through Task Scheduler or through the “schtasks” command, attackers could still abuse it if it is not disabled or removed by creating new scheduled tasks. If an administrator sees a task running under “at.exe,” this should be considered a potential red flag and investigated to determine if it is malicious in nature or if it is legitimately being used for legacy purposes. If for legacy purposes, it is recommended to move these tasks under the current schedule task function, “schtasks.exe,” where possible.

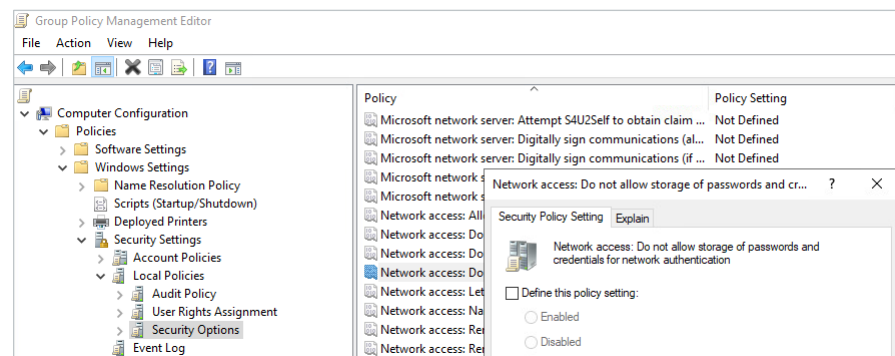
One way to protect against scheduled task abuse is to disable the use of “at.exe” to ensure that no accounts are able to use this function. One way to achieve this is by blocking executable files from running (e.g., “at.exe”) through a GPO under “Computer Configuration\Policies\Windows Settings\Security Settings\Software Restriction Policies” shown in Figure 11 above.

Restrict the Ability to Specify Alternate Credentials for Scheduled Tasks

Scheduled tasks are tied to a specific account when created. Attackers will sometimes attempt to create scheduled tasks under the SYSTEM account or attempt to specify alternate credentials for scheduled tasks to run. If these credentials are cached, they can be abused during an attack.

In order to restrict the ability for attackers to specify alternate credentials for scheduled tasks, it is recommended to ensure that ‘Network access: Do not allow storage of passwords and credentials for network authentication’ is set to ‘Enabled’ (CIS Windows 10 Enterprise Benchmark 2.3.10.4). This policy setting removes the ability for the system to save passwords or credentials for later use when it gains domain authentication. It is worth noting that this particular setting may affect some legitimate scheduled tasks. **Caution should be exercised when selecting this setting.**

Figure 14. Configuration Setting: “Network access: Do not allow storage of passwords and credentials for network authentication” is set to ‘Enabled’

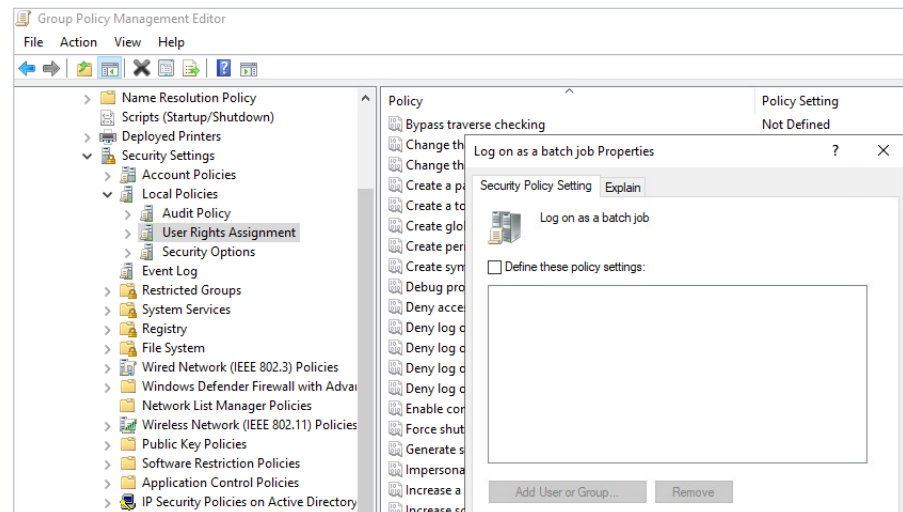


Restrict Accounts That Can Log On as a Batch Job

Task Scheduler is often used for administrative purposes. Therefore, its use should be restricted in high-security environments to prevent the misuse of system resources or to prevent attackers from launching malicious code after gaining user-level access to a system.

Enterprises can configure this setting by ensuring that 'Log on as a batch job' is set to 'Administrators' (CIS Windows 10 Enterprise Benchmark 2.2.28) (Figure 15). Note that if the enterprise is using optional components (e.g., ASP.NET, IIS⁷), it may be necessary to assign this user right to additional accounts that are required by those components. ***If they are not assigned these rights, the components may not be able to run, which can impact functionality.***

Figure 15. Configuration Setting: "Computer Configuration\Windows Settings\Security Settings\Local Policies\User Rights Assignment\Log on as a batch job" is set to 'Administrators'



⁷ (ASP.NET) Active Server Pages Network Enabled Technologies, (IIS) Internet Information Services

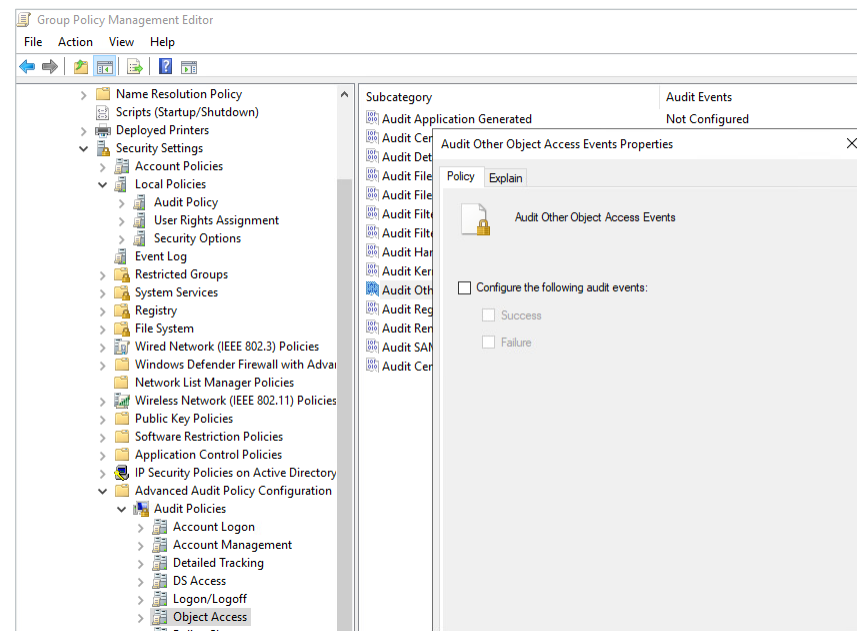
Enable Object Access Events

Along with implementing preventive measures to defend against scheduled task abuse, it is also important to log events associated with scheduled tasks in the event that an attack does occur. As with the Tarrask malware used by HAFNIUM, attackers who abuse scheduled tasks may also have the ability to hide those tasks to evade detection. Upon the creation of a scheduled task, the Windows Registry creates a key with several values, one of which is called a Security Descriptor (SD). This value, when deleted under the SYSTEM account, hides the scheduled task from both Task Scheduler and the "schtasks" command. This can reduce the visibility into potentially malicious activities occurring on the system or network.

If an administrator suspects that scheduled task abuse has occurred, one way to confirm the presence or absence of a scheduled task is to check the Windows Registry key ("HKEY_LOCAL_MACHINE\SOFTWARE\Microsoft\Windows NT\CurrentVersion\Schedule\TaskCache\Tree") for any tasks that are missing SDs.⁸ Another way to monitor for hidden tasks or any scheduled task activities is to ensure that 'Audit Other Object Access Events' is set to both 'Success and Failure' (CIS Windows 10 Enterprise Benchmark 17.6.3). This Group Policy setting can be found under "Computer Configuration\Policies\Windows Settings\Security Settings\Advanced Audit Policy Configuration\Audit Policies\Object Access\Audit Other Object Access Events."

Setting this policy allows the enterprise to audit events that are generated by Task Scheduler, including those that are created, deleted, enabled, disabled, and updated. While scheduled task creation can be a normal IT function, having this setting enabled will also track potentially malicious activity, which can be helpful during an investigation. Without this setting enabled, enterprises run the risk of missing key activities that could be indicative of an attack.

Figure 16. Configuration Setting: "Computer Configuration\Policies\Windows Settings\Security Settings\Advanced Audit Policy Configuration\Audit Policies\Object Access\Audit Other Object Access Events"



⁸ <https://www.microsoft.com/en-us/security/blog/2022/04/12/tarrask-malware-uses-scheduled-tasks-for-defense-evasion/>

As previously mentioned, all scheduled task activity is logged in either the Windows Task Scheduler Operational Event Log or Windows Security Event Log. Once the GPO setting for auditing object access events is turned on, events will begin to be logged. Below is a small portion of Event IDs that are helpful to monitor before or during an incident.

Windows > Task Scheduler > Operational Event Log

100: A scheduled task was started.

101: A scheduled task has failed.

102: A scheduled task has completed.

106: A scheduled task was registered.

107: A scheduled task triggered on scheduler.

108: A scheduled task triggered on event.

110: A scheduled task triggered by user.

111: A scheduled task was terminated.

141: A scheduled task was updated.

142: A scheduled task was deleted.

Windows > Security Event Log

4698: A scheduled task was created.

4699: A scheduled task was deleted.

4700: A scheduled task was enabled.

4701: A scheduled task was disabled.

4702: A scheduled task was updated.

Note that by default, Windows does not turn on auditing for these events. The GPO noted above must be enabled in order for the events to be logged. Additionally, it is important to keep in mind that turning on this auditing may generate a large number of events. Enterprises should ensure that ample log storage is available for storing these events and that retention policies are set. The amount of storage needed and the retention policy will differ for each enterprise.

Keep in mind that scheduled tasks can also be created using PowerShell or Windows Management Instrumentation (WMI), which may require additional logging. CIS Controls has published separate documentation for protecting against attacks that use PowerShell and/or WMI.

Mappings

As CIS is home to the CIS Critical Security Controls (CIS Controls) and CIS Benchmarks, we strive to provide readers with guidance that can be aligned to our recommendations. Additionally, CIS developed the CIS Community Defense Model (CDM) v2.0, which provides defenders prioritized best practices that defend against the top five most common attacks observed across the community. This model uses the MITRE ATT&CK (Adversarial Tactics, Techniques, and Common Knowledge) framework. Shown below are the Safeguards from the CIS Controls v8, Benchmark Recommendations from the Microsoft Windows 10 Enterprise Benchmark v1.12.0, and ATT&CK (Sub-)Techniques from MITRE ATT&CK v8.2 that can be helpful to align the defenses outlined in this document to these three sources.

CIS Critical Security Controls v8

- 4.1: Establish and Maintain a Secure Configuration Process
- 4.7: Manage Default Accounts on Enterprise Assets and Software
- 4.8: Uninstall or Disable Unnecessary Services on Enterprise Assets and Software
- 5.3: Disable Dormant Accounts
- 5.4: Restrict Administrator Privileges to Dedicated Administrator Accounts
- 6.1: Establish an Access Granting Process
- 6.2: Establish an Access Revoking Process
- 6.8: Define and Maintain Role-Based Access Control
- 8.2: Collect Audit Logs
- 8.3: Ensure Adequate Audit Log Storage

CIS Windows 10 Enterprise Benchmark v1.12.0

- 2.2.25 (L1) Ensure 'Increase scheduling priority' is set to 'Administrators, Window Manager\Window Manager Group'
- 2.2.28 (L2) Ensure 'Log on as a batch job' is set to 'Administrators'
- 2.3.10.4 (L1) Ensure 'Network access: Do not allow storage of passwords and credentials for network authentication' is set to 'Enabled'
- 17.6.3 (L1) Ensure 'Audit Other Object Access Events' is set to 'Success and Failure'

MITRE ATT&CK v8.2

- [T1053 – Scheduled Task/Job](#)
- [T1053.002 – Scheduled Task/Job: At \(Windows\)](#)
- [T1053.005 – Scheduled Task/Job: Scheduled Task](#)

Conclusion

Most cyber attacks occur due to a lack of essential cyber hygiene. Whether that be through the use and abuse of tools and techniques found on the impacted system (e.g., LotL) or through an exploited vulnerability, many attacks can be defended by implementing Safeguards found within the CIS Controls. This guide provides guidance on how an enterprise could apply the Safeguards specifically in terms of defending against or detecting abuse of scheduled tasks. By implementing the security best practices recommended in this guide, enterprises can apply a defense-in-depth strategy to strengthen their cybersecurity posture and help better defend against a scheduled task attack.

Appendix A: CIS Controls

The CIS Critical Security Controls® (CIS Controls®) are a prioritized set of actions which collectively form a defense-in-depth set of best practices that mitigate the most common attacks against systems and networks. They are developed by a community of information technology (IT) experts who apply their first-hand experience as cyber defenders to create these globally accepted security best practices. The experts who develop the CIS Controls come from a wide range of sectors, including retail, manufacturing, healthcare, education, government, defense, and others. It is important to note that while the CIS Controls address general best practices that enterprises should implement to protect their environment, some operational environments may present unique requirements not addressed by the CIS Controls or require deviations from best practices.

Implementation Groups



The number of Safeguards an enterprise is expected to implement increases based on which group the enterprise falls into.

153
TOTAL SAFEGUARDS

IG3 assists enterprises with IT security experts to secure sensitive and confidential data. IG3 aims to prevent and/or lessen the impact of sophisticated attacks.

23
SAFEGUARDS

IG2 assists enterprises managing IT infrastructure of multiple departments with differing risk profiles. IG2 aims to help enterprises cope with increased operational complexity.

74
SAFEGUARDS

IG1 is the definition of essential cyber hygiene and represents a minimum standard of information security for all enterprises. IG1 assists enterprises with limited cybersecurity expertise thwart general, non-targeted attacks.

56
SAFEGUARDS

The Implementation Group (IG) methodology was developed as a new way to prioritize the CIS Controls. These IGs provide a simple and accessible way to help enterprises of different classes focus their scarce security resources, while still leveraging the value of the CIS Controls program, community, and complementary tools and working aids.

IG1. An IG1 enterprise is small to medium-sized with limited IT and cybersecurity expertise to dedicate toward protecting IT assets and personnel. The principal concern of these enterprises is to keep the business operational, as they have a limited tolerance for downtime. The sensitivity of the data that they are trying to protect is low and principally surrounds employee and financial information. Safeguards selected for IG1 should be implementable with limited cybersecurity expertise and aimed to thwart general, non-targeted attacks. These Safeguards will also typically be designed to work in conjunction with small or home office commercial off-the-shelf (COTS) hardware and software.

IG2. An IG2 enterprise employs individuals responsible for managing and protecting IT infrastructure. These enterprises support multiple departments with differing risk profiles based on job function and mission. Small enterprise units may have regulatory compliance burdens. IG2 enterprises often store and process sensitive client or enterprise information and can withstand short interruptions of service. A major concern is loss of public confidence if a breach occurs. Safeguards selected for IG2 help security teams cope with increased operational complexity. Some Safeguards will depend on enterprise-grade technology and specialized expertise to properly install and configure.

IG3. An IG3 enterprise employs security experts that specialize in the different facets of cybersecurity (e.g., risk management, penetration testing, application security). IG3 assets and data contain sensitive information or functions that are subject to regulatory and compliance oversight. An IG3 enterprise must address availability of services and the confidentiality and integrity of sensitive data. Successful attacks can cause significant harm to the public welfare. Safeguards selected for IG3 must abate targeted attacks from a sophisticated adversary and reduce the impact of zero-day attacks.

If you would like to know more about the Implementation Groups and how they pertain to enterprises of all sizes, there are many resources that explore the Implementation Groups and the CIS Controls in general on our website at <https://www.cisecurity.org/controls/cis-controls-list/>.

Appendix B: Links and Resources

CIS Critical Security Controls (CIS Controls) v8:

Learn more about the CIS Controls, including how to get started, why each Control is critical, procedures and tools to use during implementation, and a complete listing of Safeguards for each Control.

CIS Controls Assessment Specification: Provides an understanding of what should be measured in order to verify that the Safeguards are properly implemented.

CIS Controls Navigator: Learn more about the Controls and Safeguards and see how they map to other security standards (e.g., Cybersecurity Maturity Model Certification (CMMC); National Institute of Standards and Technology Standard Publication (NIST SP) 800-53 Rev. 5; Payment Card Industry Data Security Standard (PCI DSS); MITRE ATT&CK).

CIS Controls Self Assessment Tool (CIS CSAT):

Enables enterprises to assess and track their implementation of the CIS Controls for Versions 8 and 7.1.

CIS Community Defense Model (CDM) v2.0: A guide published by CIS that leverages the open availability of comprehensive summaries of attacks and security incidents, and the industry-endorsed ecosystem that is developing around the MITRE ATT&CK Framework.

CIS Risk Assessment Method (CIS RAM) v2.1:

An information security risk assessment method that helps enterprises implement and assess their security posture against the CIS Controls.

CIS SecureSuite Membership: Membership with access to CIS-CAT (Configuration Assessment Tool) Pro Assessor, CIS Build Kits, CIS Benchmarks, and more.

CIS Benchmarks: Secure configuration guidelines for 100+ technologies, including operating systems, applications, and network devices.

CIS-CAT Pro: Tool to scan for proper CIS Benchmark configurations for applications, operating systems, and network devices.

CIS Build Kits: ZIP files that contain a GPO for each profile within the corresponding CIS Benchmark.

CIS Hardened Images®: Virtual machine images securely pre-configured to the CIS Benchmarks.

CIS WorkBench: Get involved in one of our many communities.

The Center for Internet Security, Inc. (CIS®) makes the connected world a safer place for people, businesses, and governments through our core competencies of collaboration and innovation. We are a community-driven nonprofit, responsible for the CIS Critical Security Controls® and CIS Benchmarks™, globally recognized best practices for securing IT systems and data. We lead a global community of IT professionals to continuously evolve these standards and provide products and services to proactively safeguard against emerging threats. Our CIS Hardened Images® provide secure, on-demand, scalable computing environments in the cloud.

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